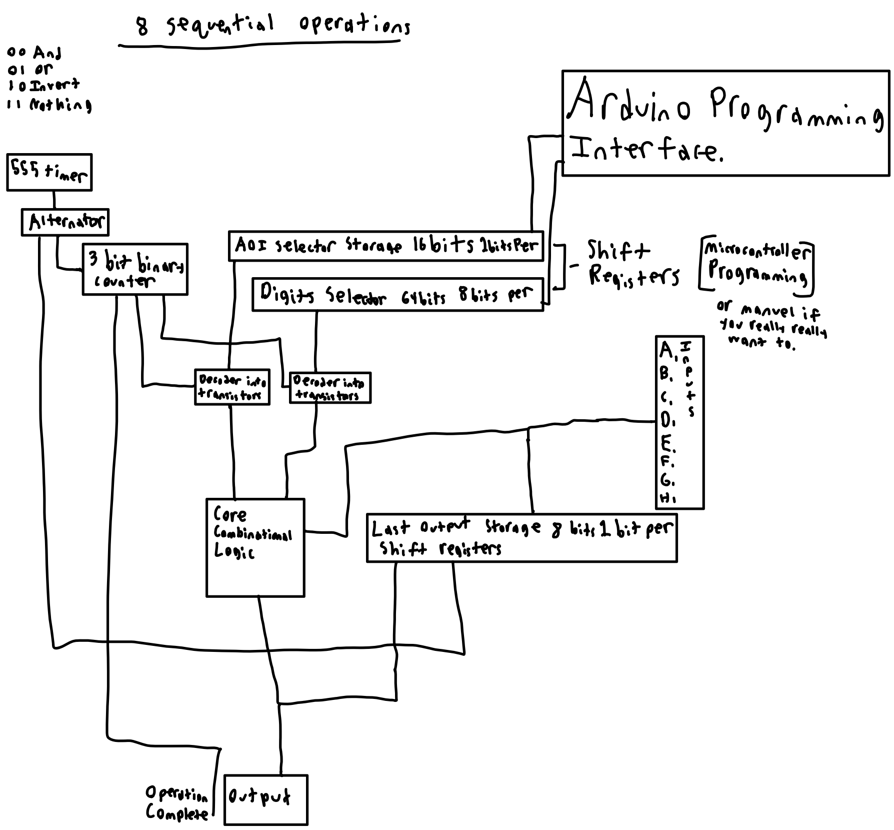
Problem Statement: A single use state machine is wasteful. Why not have a programmable state machine?

Design Statement: Design and build an programmable logic device that can perform 8 AOI operations on a set of inputs, and can be programmed via a microcontroller.

Criteria/Constraints:
A 555 timer will be used to drive alternator, which will drive a 3 bit binary counter. The program storage is stored on shift registers so it can be programmed via a microcontroller (Or by hand if you really want to!). There will be 8 inputs and 1 output.
The clock frequency will be very high. ~1,000 Hz. The faster it is, the faster the PLD will run (Quicker response time).
There will be a “run” button run the “code.”
The circuit will be in an enclosure that displays its inner workings, maybe in a multi level setup so it takes up less space.

Block Diagram



Truth table for the core combinational logic

AOI Logic

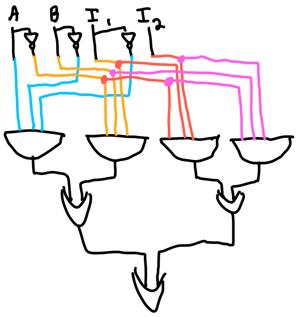
A	B	I_1	I_2	K
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

AB
00 And
01 or
10 Invert
11 Nothing

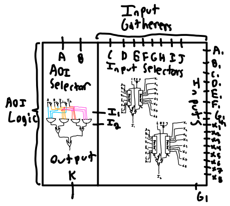
	I_1I_2	$\bar{I}_1I_2$	$\bar{I}_1\bar{I}_2$	$I_1\bar{I}_2$
AB	0	0	0	0
$\bar{A}B$	1	1	0	1
$\bar{A}\bar{B}$	1	0	0	0
$A\bar{B}$	0	1	1	0

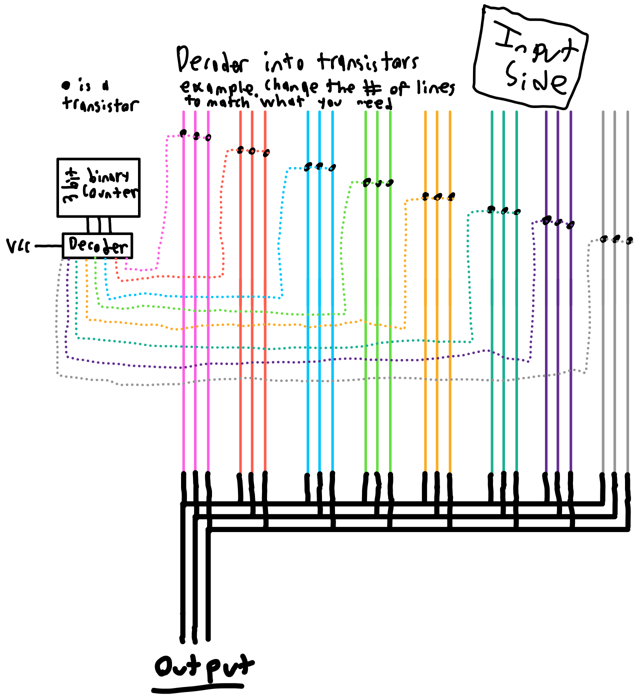
$K = A\bar{B}\bar{I}_1 + \bar{A}BI_1 + \bar{A}I_1I_2 + \bar{A}BI_2$

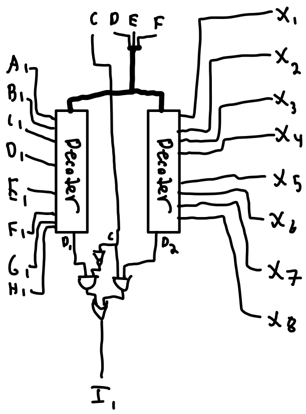
- Use 3 input and gates

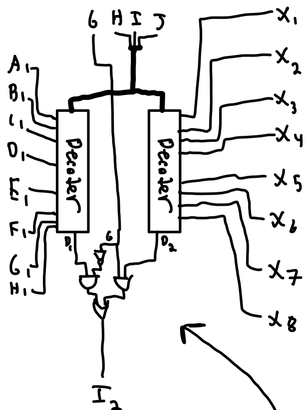


Core Combinational Logic









Same with this circuit, just replace I_1 with I_2 and C with G.

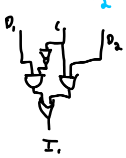
C	D_1	D_2	I_1
0	0	0	0
1	0	0	0
0	1	0	1
1	1	0	0
0	0	1	0
1	0	1	1
0	1	1	1
1	1	1	1

$I_1 = \bar{C}D_1\bar{D}_2 + C\bar{D}_1D_2 + \bar{C}D_1D_2 + CD_1D_2$

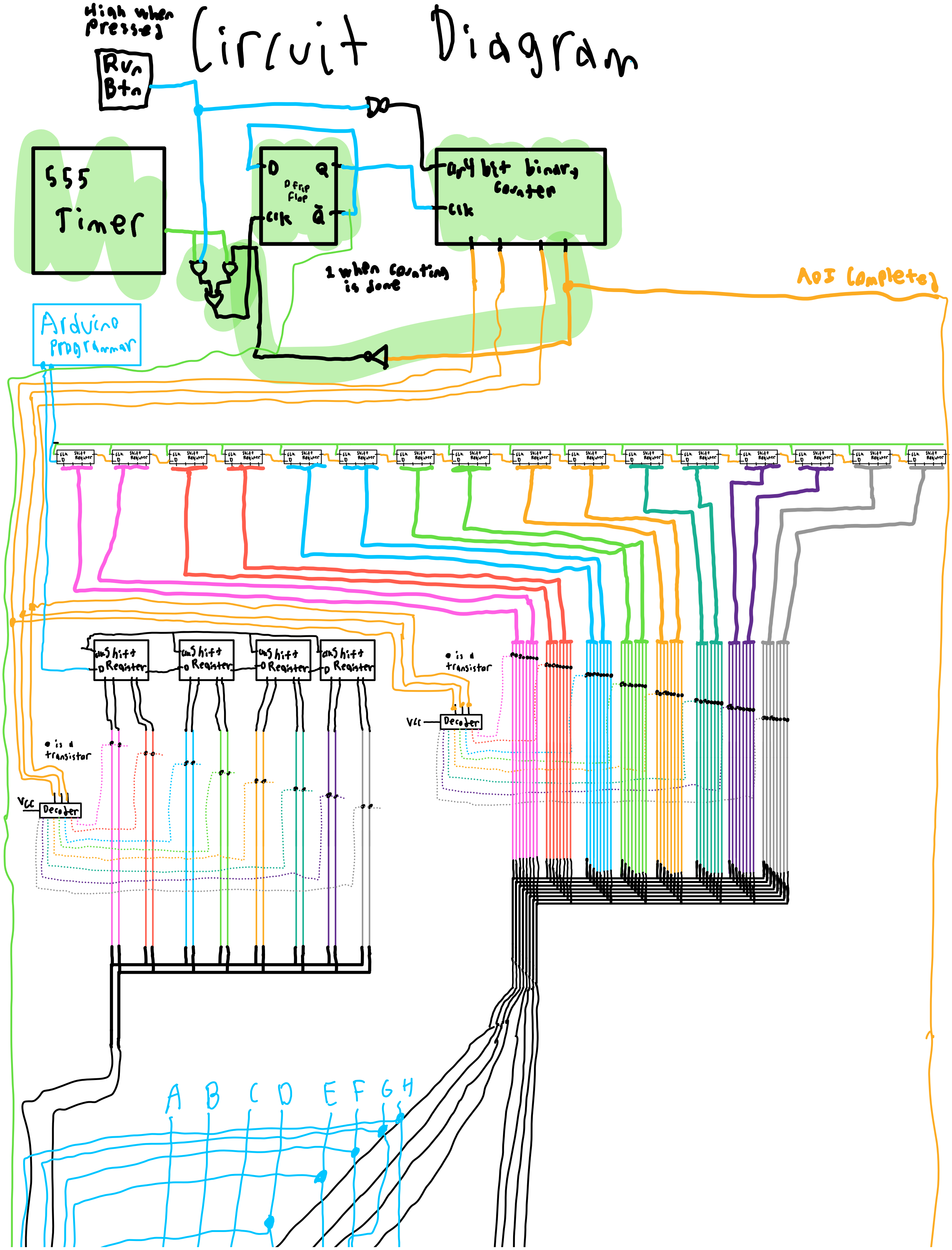
$I_1 = \bar{C}D_1(D_2+\bar{D}_2) + CD_2(D_1+\bar{D}_1)$

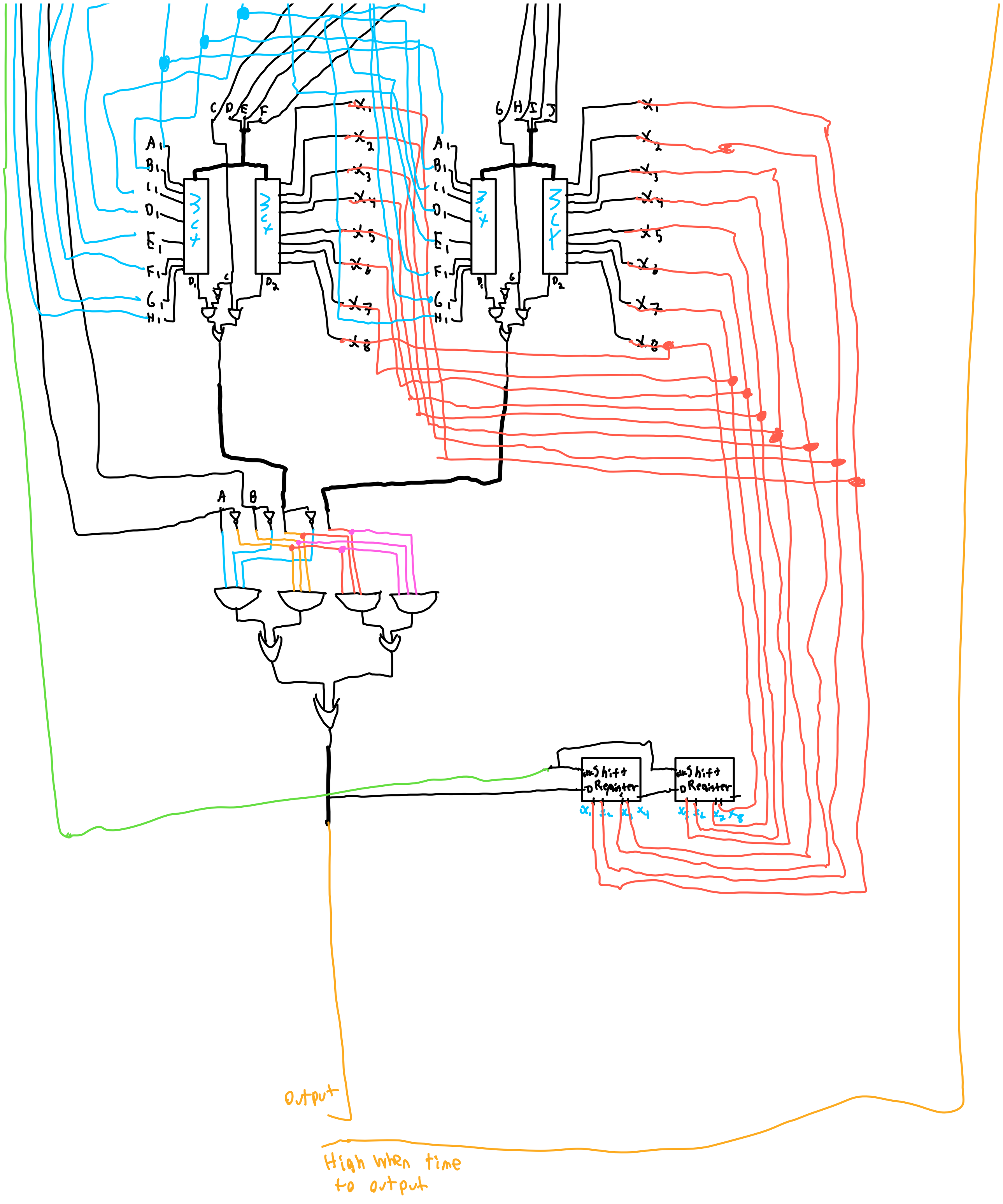
$I_1 = \bar{C}D_1(1) + CD_2(1)$

$I_1 = \bar{C}D_1 + CD_2$



Circuit Diagram





C D E F
G H I J
A1 B1 C1 D1 E1 F1 G1 H1
X1 X2 X3 X4 X5 X6 X7 X8
A B
Shift Register
Shift Register
output
High when time to output

AOI Logic

A	B	I_1	I_2	K
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

AB
00 And
01 or
10 Invert
11 Nothing

	I_1I_2	$\bar{I_1}I_2$	$\bar{I_1}\bar{I_2}$	$I_1\bar{I_2}$
AB	0	0	0	0
$\bar{A}B$	1	1	0	1
$\bar{A}\bar{B}$	1	0	0	0
$A\bar{B}$	0	1	1	0

$K = A\bar{B}\bar{I_1} + \bar{A}BI_1 + \bar{A}I_1I_2 + \bar{A}BI_2$

- Use 3 input and gates